Language-Guided and Similarity-Aware Network for Few-Shot Classification of Coastal Wetland Hyperspectral Images

Qiaoli Zhang, Jiangtao Peng, *Senior Member, IEEE*, and Weiwei Sun, *Senior Member, IEEE*

***Abstract*—In recent years, few-shot learning (FSL) has made significant progress in hyperspectral image classification (HSIC) by transferring meta-knowledge from a source domain with sufficient labeled samples to a target domain with limited labeled samples. However, existing FSL methods face two key challenges in coastal wetland HSIC applications, i.e., prototype instability due to limited labeled samples and domain shift due to the domain's distribution difference. These challenges are further exacerbated by the unique characteristics of coastal wetland environments, which have complex land cover classes and subtle spectral differences between land cover classes. To address these limitations, we propose a language-guided and similarity-aware network (LGSAnet) for few-shot coastal wetland HSIC in this article. The network mainly consists of two modules: language-guided prototype alignment (LGPA) and similarity-aware prototype calibration (SAPC). The LGPA module uses linguistic features to guide the learning of visual features, enabling the model to obtain visual representations enriched with linguistic prior knowledge. This guided alignment can learn more accurate and stable visual prototype representations, thereby improving the accuracy of the model. To mitigate domain shift, an SAPC module is constructed to refine the target domain prototypes and minimize domain-specific differences using the domain's semantic similarity. The experimental results on three coastal wetland hyperspectral datasets demonstrate that the proposed LGSAnet outperforms existing state-of-the-art FSL methods.**

***Index Terms*—Coastal wetland, few-shot learning (FSL), hyperspectral image classification (HSIC), text feature.**

I. Introduction

COASTAL wetlands play a vital role in global ecosystems and serve as important buffers between terrestrial and marine environments. These ecosystems provide various environmental benefits, such as mitigating coastal erosion, improving water quality, and supporting biodiversity. However, coastal wetlands are increasingly vulnerable to degradation due to human destruction and climate change, so monitoring and protecting coastal wetlands is essential [1], [2]. Hyperspectral images (HSIs) provide important support for environmental monitoring, which can capture detailed spectral information in many continuous bands, thereby revealing subtle differences in land cover classes [3], [4], [5], [6]. This makes HSI classification (HSIC) an effective way to map and analyze the diverse ecosystems of coastal wetlands. In the early stages of HSIC, traditional classifiers [7], [8], [9] typically used morphological profiles [10], Gabor features [11], or other spatial–spectral features for classification. Although these classifiers have achieved good performance, their performance heavily relies on the raw spectral information or hand-crafted features.

Over the past decade, many deep-learning-based methods have been introduced for HSIC, as they demonstrate excellent ability in automatically mining task-relevant spatial–spectral features of HSIs [12], [13], [14], [15]. Chen et al. [16] proposed a stacked autoencoder (SAE) method to extract deep spatial–spectral features. Since SAE-based methods involve a large number of parameters, they further proposed to use the convolutional neural networks (CNNs) for HSIC [17]. Following the success of CNNs, many variants have been proposed subsequently [18], [19], [20], [21], [22]. However, as the depth of the network increases, CNN-based methods may suffer from gradient explosion or vanishing. To address this issue, Zhong et al. [23] proposed a spectral–spatial residual network (SSRN). Roy et al. [24] proposed a hybrid spectral network (HybridSN) by combining the 3D-CNN and 2D-CNN, which balances computational complexity and classification performance. Despite their effectiveness in learning discriminative features, CNN-based models generally require a large amount of labeled data for training [25], [26], [27].

Due to the high cost of labeling land cover pixels, labeled samples are usually limited, which degrades the performance of CNN-based methods. To address the challenge of small-sample classification, few-shot learning (FSL) has been developed [28], [29]. Inspired by the human visual system's ability to quickly recognize new categories from limited labeled samples, FSL enables accurate predictions with few labeled samples. Recently, FSL has also been applied to HSIC [30], [31], [32], [33], [34], [35], [36]. Li et al. [37] proposed a deep cross-domain FSL (DCFSL) method for HSIC that mitigates domain shift through conditional adversarial domain adaptation. Zhang et al. [38] used graph neural networks to alleviate domain shift and proposed a graph information

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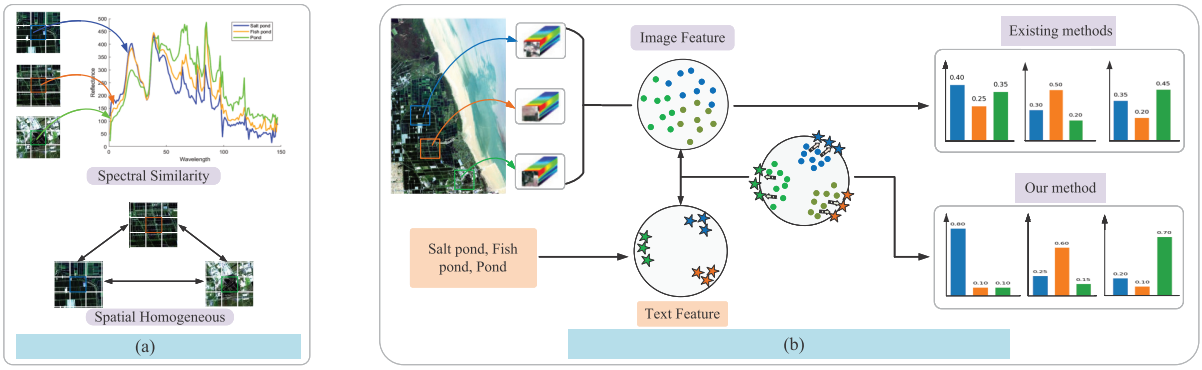


Fig. 1. Motivation of the LGPA. (a) Challenges for HSIC, i.e., spectral similarity and spatial homogeneous between different categories. (b) Existing methods primarily rely on visual features and ignore textual features, which leads to the dispersion of features within the same category and confusion between features of different categories. Our method integrates textual features to guide the visual features, thereby facilitating improved clustering of the same category.

aggregation cross-domain FSL (Gia-CFSL) method. Liu et al. [39] introduced a refined prototype contrastive learning FSL (RPCL-FSL) method, which integrates contrastive learning and FSL into an end-to-end network for HSIC. Li et al. [40] proposed a few-shot classification framework with self-supervised learning (FSCF-SSL), which incorporates self-supervised learning into the FSL framework and uses natural images for cross-domain transfer learning. Li et al. [41] proposed a lightweight self-pooling transformer (SPFormer) for few-shot HSIC, combining a self-supervised autoencoder with a parameter-free spectral-pixel interaction module to achieve efficient feature extraction.

Although the aforementioned FSL methods have demonstrated promising performance, they rely solely on visual features. As illustrated in Fig. 1(b), existing HSIC approaches typically construct visual representations by combining pixel-level spectral features with local spatial features, which may be difficult to distinguish spectral-similar categories. The main reason is shown in Fig. 1(a). In HSIs, different land cover types may share highly similar spectral and spatial characteristics. For instance, salt ponds, fish ponds, and eutrophic water bodies exhibit very similar spectral curves, and it is very likely to produce substantial false positives.

Inspired by the fact that humans naturally integrate information from multiple modalities such as vision and language, many studies have explored the use of multimodal information to enhance classification performance [42], [43]. Radford et al. [44] proposed a contrastive language–image pretraining (CLIP) model, which learns instance-level associations between images and textual descriptions through a contrastive learning framework. Building on this idea, Chen et al. [45] proposed an adaptive mechanism to combine visual and linguistic priors, while Han et al. [46] designed a textual-enhanced meta-fusion module to align visual and semantic features for FSL. Li et al. [47] further proposed a knowledge-guided semantic transfer network that facilitates category-level relationship propagation across modalities. However, most of these methods either focus on instance-level alignment or fuse multimodal information at the feature level, without explicitly addressing prototype instability caused by limited data in few-shot HSIC tasks. To overcome this limitation, we propose a novel language-guided prototype alignment (LGPA) framework for coastal wetland HSIC, as illustrated in Fig. 1(b). LGPA constructs category-level semantic prototypes and directly aligns them with spatial–spectral visual prototypes through a cross-modal contrastive objective. By introducing higher order semantic constraints at the prototype level, LGPA effectively stabilizes prototype distributions and enhances discriminative feature learning, particularly for distinguishing spectrally similar but semantically different land cover classes in few-shot scenarios.

On the other hand, to address the domain shift problem in few-shot HSIC, traditional domain adaptation methods such as adversarial learning [48] and contrastive learning [49] have been widely explored. Adversarial learning approaches enforce a global alignment of feature distributions between domains via domain classifiers, which may blur class boundaries. Contrastive learning methods rely on instance-level similarity and often lack explicit class supervision. Here, we introduce a similarity-aware prototype calibration (SAPC) module to directly model interclass cross-domain relationships through a prototype similarity matrix. Instead of aligning global feature distributions or instance pairs, SAPC dynamically adjusts each target prototype based on its semantic similarity to source prototypes, which can preserve domain-specific features and avoiding negative transfer.

In summary, we propose a language-guided and similarity-aware network (LGSAnet) to enhance the prototype representations and reduce the domain shift for cross-domain few-shot coastal wetland HSIC. The proposed network integrates cross-modal information and uses a similarity-aware mechanism to improve the classification performance. First, it uses text descriptions as language modality information to guide the learning of visual prototypes. By aligning language and visual prototypes, it can enrich the semantic representation of visual features. Second, a similarity-aware mechanism is designed to alleviate the distribution differences between the source and target domains. It dynamically adjusts and calibrates target prototype representations to mitigate domain shift. The main contributions are as follows.

1) We propose an LGSAnet for few-shot coastal wetland HSIC. It uses language prior knowledge as supervisory

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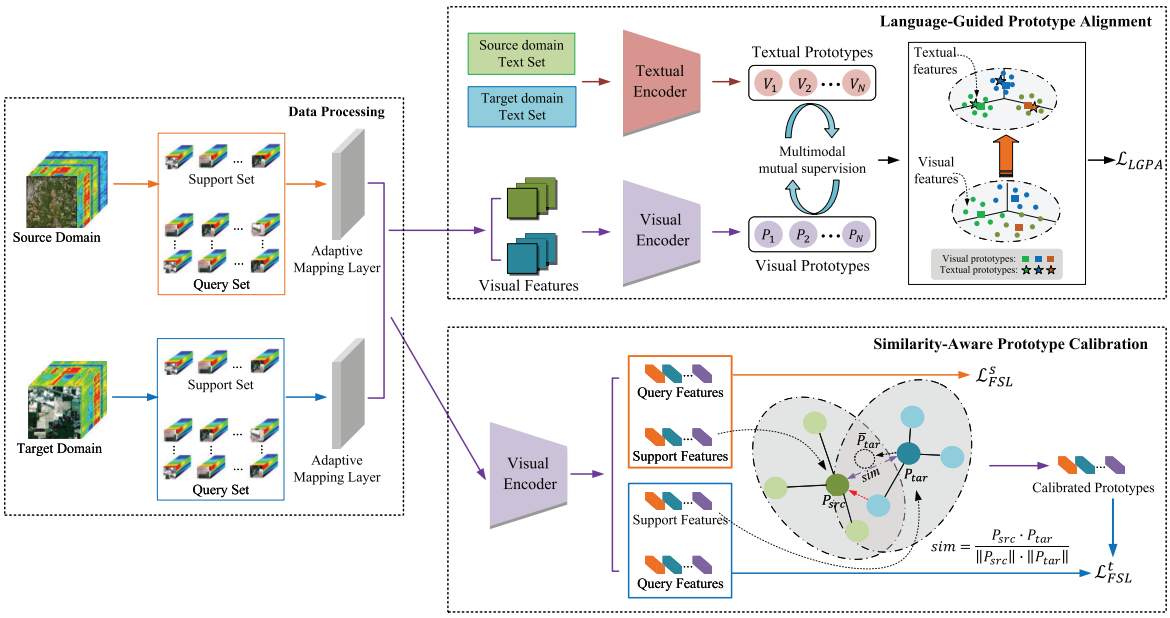


Fig. 2. Flowchart of the proposed LGSAnet. It mainly consists of two parts, i.e., the LGPA module and SAPC module.

information to enhance the visual feature learning, thus improving the model's stability and effectiveness.

2) A language-guided alignment method is constructed to align visual and linguistic prototypes within the multimodal embedding space, allowing the visual encoder to integrate semantic knowledge from text descriptions and enhance the prototype representations of the visual modality.
3) A similarity-aware module is designed to dynamically calibrate the target domain prototypes using similarity information between the source and target domains, thereby reducing domain shift and improving cross-domain classification accuracy.

The rest of this article is organized as follows. Section II describes the proposed LGSAnet. Section III shows the experimental results and analysis. Finally, Section IV concludes this article.

II. Proposed Method

A. LGSAnet Framework

Fig. 2 illustrates the flowchart of the proposed LGSAnet, which primarily consists of data preprocessing, LGPA module, SAPC module, and FSL in the source and target domains. Initially, FSL is performed on the source domain to learn general feature representations and category prototypes. To enhance the stability and accuracy of prototypes, the LGPA module is designed to incorporate cross-modal information from both visual and textual modalities. To mitigate the distribution discrepancy between domains, the SAPC module is constructed to align the visual prototypes between the source and target domains by fusing stable source domain prototypes with biased target domain prototypes.

B. Data Processing

Assume that the source domain HSI dataset D_s contains C_s classes, and the target domain HSI dataset D_t has C_t classes, where $C_s \geq C_t$ and $D_s \cap D_t = \emptyset$. For FSL, all samples in the source domain are labeled, whereas only a few labeled samples are available in the target domain (e.g., five labeled samples per class in this study). In the source domain, 200 labeled samples per class are randomly selected for meta-training. To expand the labeled samples in the target domain, Gaussian noise is added to the existing labeled samples for data augmentation, thereby increasing the number of labeled samples per class from 5 to 200. For a given training sample x, we can synthesize new augmented samples x' through the following operation:

$$x' = f(x) + \lambda \cdot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \tag{1}$$

where $f(\cdot)$ represents a linear transformation function with a randomly sampled scaling factor $\gamma \sim \mathcal{U}(0.9,\ 1.1)$ and $\lambda = 0.04$ regulates the intensity of random Gaussian noise ε [40].

The support and query sets are constructed in the same way for both the source and target domains. For the source domain, an N-way K-shot setup is used for episodic training. In each meta-task, N categories are randomly selected, and K samples from each category are used to form the support set, while K_q samples per class are selected to form the query set. We denote the support and query sets of source domain as: $S_s = \{(x_i^{ss}, y_i^{ss})\}_{i=1}^{N\times K}$, $Q_s = \{(x_j^{sq}, y_j^{sq})\}_{j=1}^{N\times K_q}$. In this article, $N = C_t$, $K = 1$, and $K_q = 19$.

C. Visual Encoder

To mine spatial and spectral information of HSIs, a deep 3-D residual visual encoder is used to extract spatial-spectral features. To fully use the meta-knowledge learned from the source domain, the visual encoder's weights are shared between the source and target domains, and an adaptive mapping layer is used to map the source and target domains to the same dimension. Specifically, the adaptive mapping layer is a 2-D convolution operation with d convolution kernels of size

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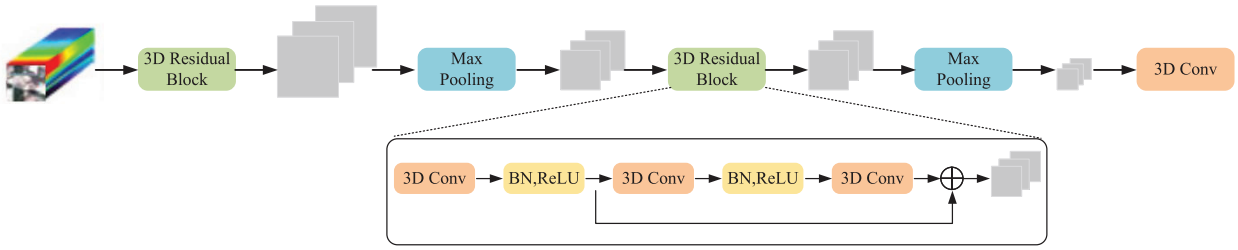


Fig. 3. Flowchart of visual encoder.

$1 \times 1 \times$ ch, where d represents the output data dimension and ch represents the input data dimension.

After the adaptive mapping layer, a spectral–spatial residual encoder is introduced to extract spatial–spectral features. As shown in Fig. 3, the visual encoder consists of two 3-D residual blocks [38], two pooling layers, and one convolutional layer. Each residual block contains three convolution kernels of size $3 \times 3 \times 3$ and is connected to a 3-D max-pooling layer to reduce computational complexity.

D. Textual Encoder

To capture the semantic relationships of land cover categories [44], a textual encoder based on a modified language transformer is designed. Similar to the CLIP model, the text is represented using lowercase byte-pair encoding with a vocabulary size of 49152. The extracted features are normalized at each layer and then linearly projected into a semantic space to facilitate effective alignment with visual features during classification. A three-layer stacked transformer encoder is used for feature encoding. Due to the reduced transformer depth compared with the original CLIP model, we perform partial weight loading by selectively transferring matching parameters and fine-tune the entire textual encoder during training. This strategy effectively transfers generalizable language knowledge while maintaining flexibility for adaptation to hyperspectral land cover classification tasks under few-shot conditions.

In addition, to incorporate prior knowledge about the categories from the source and target domains, a fill-in-the-blank approach is used to create concise text descriptions for each land cover category. For example, if the label of the land cover category is "Forest," the description is transformed into "An HSI of Forest."

E. Source Domain FSL

FSL is performed on both the source and target domains. Meta-knowledge is transferred from the source domain to the target domain through a shared-weight visual encoder, which ensures effective adaptation to new categories with limited labeled data.

In the prototypical network, prototypes are obtained by averaging the features of support set samples from each class. Taking FSL in the source domain as an example, the visual prototype for the cth category in the support set of source domain is computed as follows:

$$P_c^s = \frac{1}{K} \sum_{x_i^{ss} \in S_c^s} f_\theta\left(x_i^{ss}\right), \quad c = 1, \ldots, N \tag{2}$$

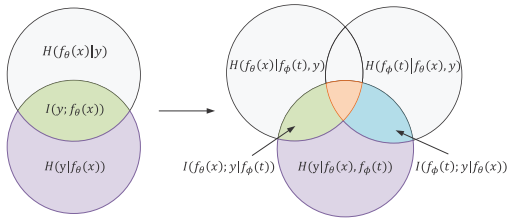


Fig. 4. Information diagrams in LGPA. (Left diagram) Scenario where only visual information is used for classification. In this case, the remaining uncertainty of the target y, given the visual features $f_\theta(x)$, is measured by the conditional entropy $H(y|f_\theta(x))$. This reflects the model's uncertainty when relying solely on visual features. (Right diagram) Effect of incorporating textual information to visual features. Here, the remaining uncertainty of y is represented by $H(y|f_\theta(x), f_\phi(t))$. Since textual feature provides complementary discriminative information, it significantly reduces the conditional entropy, indicating that the model obtains more stable prototypes.

where S_c^s denotes the set of samples belonging to the cth category in the support set of source domain, and f_θ is the visual feature extractor.

The samples from the query set Q_s are passed through the feature encoder to obtain their feature representations. For a query sample $x_j^s \in Q_s$ with a true label c_j, the prototypical network can predict the probability that the sample x_j^s belongs to class c_j as follows:

$$p\left(y_j = c_j | x_j^{ss}\right) = \frac{\exp\left(-d\left(f_\theta\left(x_j^{ss}\right), P_{c_j}^s\right)\right)}{\sum_{c=1}^{N} \exp\left(-d\left(f_\theta\left(x_j^{ss}\right), P_c^s\right)\right)} \tag{3}$$

where $d(\cdot)$ represents the Euclidean distance. The loss for each episode during the training phase can be calculated as follows:

$$\mathcal{L}_{\mathrm{FSL}}^s = -\frac{1}{N \times K_q} \sum_{j=1}^{N \times K_q} \log p\left(y_j = c_j | x_j^{ss}\right). \tag{4}$$

F. Language-Guided Prototype Alignment

The sparsity of samples in few-shot HSIC introduces two challenges: the bias of visual features and instable category prototype representations. These issues fundamentally stem from reduction of mutual information $I(y; f_\theta(x))$ between visual features and labels under the few-shot conditions. As shown in the left diagram of Fig. 4, when $I(y; f_\theta(x))$ decreases, the conditional entropy $H(y|f_\theta(x))$ increases accordingly. This reflects the increase in label uncertainty given the visual features, which severely reduces the prototype reliability. To mitigate this uncertainty, we introduce textual features as semantic priors through information gain $\mathrm{IG} = H(y|f_\theta(x)) - H(y|f_\theta(x), f_\phi(t))$, which reduces joint

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conditional entropy $H(y|f_\theta(x), f_\phi(t))$ by aligning cross-modal semantics.

For each episode, the label texts of the source and target domains are denoted as T_s and T_t, respectively. Taking the source domain as an example, for each class $c \in C_s$, the corresponding label text $t_c^s \in T_s$, the linguistic prototype of the cth class is expressed as follows:

$$V_c^s = \frac{1}{K} \sum_{t_i^s \in t_c^s} f_\phi\left(t_i^s\right) \tag{5}$$

where f_ϕ is the text feature extractor.

For the N-way one-shot meta-task, the visual and linguistic prototype sets are represented as $P^s = \{P_1^s, P_2^s, P_3^s, \ldots, P_N^s\}$ and $V^s = \{V_1^s, V_2^s, V_3^s, \ldots, V_N^s\}$, where P_i^s and V_i^s are the visual and linguistic prototypes of the same class, which form positive sample pairs, while other pairs are considered as negative samples. The contrastive loss between P_i^s and V_i^s can be defined as follows:

$$\ell_{\mathrm{cl}}^s\left(P_i^s, V_i^s\right) = -\log\left(\frac{\exp\left(\frac{\mathrm{sim}\left(P_i^s, V_i^s\right)}{\tau}\right)}{\sum_{k=1,k\neq i}^{2N} \exp\left(\frac{\mathrm{sim}\left(P_i^s, V_k^s\right)}{\tau}\right)}\right) \tag{6}$$

where sim(.) denotes the similarity measurement function, and τ is a temperature coefficient ($\tau = 0.5$ in the experiments). According to the InfoNCE bound [50], the contrastive loss $\ell_{\mathrm{cl}}^s(P_i^s, V_i^s)$ provides a lower bound on the mutual information $I(P_i^s; V_i^s)$, i.e., $I(P_i^s; V_i^s) \geq \log M - \ell_{\mathrm{cl}}^s$., where M is the number of samples in a batch. It is clear that minimizing the contrastive loss $\ell_{\mathrm{cl}}^s(P_i^s, V_i^s)$ effectively maximizes the mutual information $I(P_i^s; V_i^s)$, and hence reduces the conditional entropy $H(y \mid f_\theta(x), f_\phi(t))$. It can lead to a more stable and discriminative category prototype representation.

The LGPA loss for the source domain can be expressed as follows:

$$\mathcal{L}_{\mathrm{LGPA}}^s = \frac{1}{2N} \sum_{i=1}^{N} \left(\ell_{\mathrm{cl}}^s\left(P_i^s, V_i^s\right) + \ell_{\mathrm{cl}}^s\left(V_i^s, P_i^s\right)\right). \tag{7}$$

For each episode, the LGPA loss for both the domains can be expressed as follows:

$$\mathcal{L}_{\mathrm{LGPA}} = \mathcal{L}_{\mathrm{LGPA}}^s + \mathcal{L}_{\mathrm{LGPA}}^t. \tag{8}$$

G. Similarity-Aware Prototype Calibration

To address the domain shift problem inherent in cross-domain few-shot classification, we propose an SAPC module. This module leverages semantic relationships between the source and target domain prototypes to dynamically refine target prototypes, thereby improving their alignment with true category semantics. During meta-training in the source domain, sufficient labeled samples allow the model to learn semantically meaningful and stable visual prototypes. These source prototypes serve as a semantic reference space to guide the calibration of target domain prototypes. As shown in Fig. 5, the refined target prototype is computed as follows:

$$\bar{P}_c^t = \alpha P_c^t + (1-\alpha)\,\Delta P_c^t \tag{9}$$

where P_c^t is the original prototype of the cth class in the target domain, ΔP_c^t denotes the calibration deviation derived from source domain prototypes, and $\alpha \in [0, 1]$ balances the retention of original target information and the degree of semantic alignment.

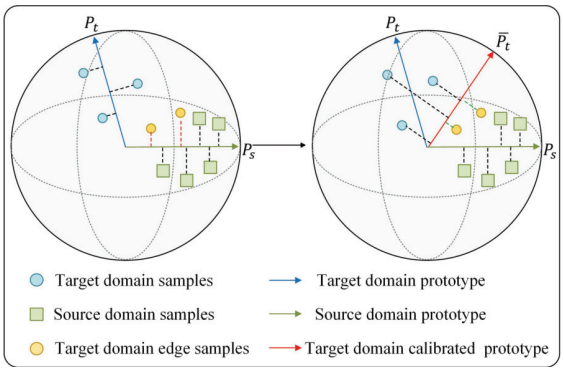


Fig. 5. Illustration of the SAPC. The blue circles and green boxes represent target and source samples, respectively. The dotted line indicates the distance from the sample to the prototype. (Left) There are two target samples denoted as yellow circles that are closer to the source prototype. (Right) After calibrating the target domain prototype, these two yellow circles are closer to the target domain prototype.

Considering the category discrepancy between the source and target domains, we first compute the cosine similarity between their prototypes to capture potential semantic correspondences

$$\mathrm{sim}_{c_s,c_t} = \frac{P_{c_s}^s \cdot P_{c_t}^t}{\left\|P_{c_s}^s\right\| \cdot \left\|P_{c_t}^t\right\|}. \tag{10}$$

Subsequently, the calibration deviation is defined as a similarity-weighted aggregation of source prototypes

$$\Delta P_{c_t}^t = \sum_{c_s=1}^{N} \frac{e^{\mathrm{sim}_{c_s,c_t}}}{\sum_{i=1}^{N} e^{\mathrm{sim}_{i,c_t}}} \cdot P_{c_s}^s. \tag{11}$$

This formulation allows each target prototype to be softly guided by multiple related source prototypes, ensuring its position in a semantically coherent subspace. Consequently, the calibrated target domain prototype is given by

$$\bar{P}_{c_t}^t = \alpha P_{c_t}^t + (1-\alpha)\,\Delta P_{c_t}^t. \tag{12}$$

By integrating guidance from semantically relevant source prototypes, SAPC effectively reduces intraclass variance and aligns cross-domain categories in the feature space. This helps mitigate prototype instability and domain shift, especially under few-shot scenarios with limited target data.

The FSL loss $\mathcal{L}_{\mathrm{fsl}}^t$ for each episode in the target domain is similar to that of the source domain (4). Consequently, the total loss for FSL can be formulated as follows:

$$\mathcal{L}_{\mathrm{FSL}} = \mathcal{L}_{\mathrm{FSL}}^s + \mathcal{L}_{\mathrm{FSL}}^t. \tag{13}$$

In the proposed algorithm, both the visual and textual encoders are trained jointly. By combining the aforementioned loss functions, the overall loss of the network can be expressed as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{FSL}} + \mathcal{L}_{\mathrm{LGPA}}. \tag{14}$$

The detail of the training procedure of the LGSAnet method is shown in Algorithm 1.

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Algorithm 1 LGSAnet Training Process

Input: The visual feature extractor f_θ
The textual feature extractor f_ϕ
The support set S_s and query set Q_s in D_s
The support set S_t and query set Q_t in D_t
The source domain label text set T_s
The target domain label text set T_t
Number of training episodes E
Output: The loss of model

if episode $< E$ **do**
  Train with S_s, Q_s, S_t, Q_t, T_s, and T_t
  Calculate visual prototypes P_c^s by (2)
  Calculate textual prototypes V_c^s by (5)
  Calculate P_c^t same as P_c^s
  Calculate V_c^t same as V_c^t
  for $x_i^{ss} \in S_s$, $x_j^{sq} \in Q_s$ **do**
    Calculate $\mathcal{L}_{\text{FSL}}^s$ of an episode by (4)
  end for
  for $x_i^{ss} \in S_s$, $t_i^s \in T_s$ **do**
    Calculate $\mathcal{L}_{\text{LGPA}}^s$ of an episode by (7)
  end for
  Calculate $\mathcal{L}_{\text{LGPA}}^t$ same as $\mathcal{L}_{\text{LGPA}}^s$
  Calculate $\mathcal{L}_{\text{LGPA}}$ of an episode by (8)
  for $x_i^{ss} \in S_s$, $x_i^{ts} \in S_t$ **do**
    Calculate $\bar{P}_{c_t}^t$ by (12)
    Calculate $\mathcal{L}_{\text{FSL}}^t$ of an episode same as $\mathcal{L}_{\text{FSL}}^s$
  end for
  Calculate $\mathcal{L}_{\text{FSL}}$ of an episode by (13)
  Return $\mathcal{L}_{\text{total}}$ of an episode by (14)
end if

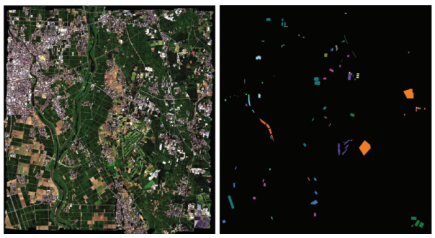


Fig. 6. Pseudocolor composite image and ground-truth map of Chikusei.

III. Experimental Results and Analysis

A. Datasets

In the experiments, we adopt the Chikusei (CH) dataset as the source domain D_s and use the Huanghekou (HHK), Yancheng (YC), and Hangzhouwan (HZW) coastal wetlands hyperspectral datasets as the target domain D_t, respectively.

1) Chikusei (CH): This dataset was captured using the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan. It contains 128 spectral bands covering the wavelength range from 343 to 1018 nm, with a spatial resolution of 2.5 m. The image size is 2517 × 2335, and the dataset includes 19 categories. Detailed information is provided in Fig. 6 and Table I.

2) Huanghekou (HHK): This hyperspectral dataset was captured over the Huanghekou (i.e., Yellow River Estuary) wetlands using the GF-5 sensor in 2019. It contains 285 spectral bands spanning the wavelength range from 390 to 2513 nm, with an image size of 1185 × 1342 pixels and a spatial resolution of 30 m. The dataset originally included 21 categories, but categories 20 (Tamarix) and 21 (Intertidal saltwater) were removed, while categories 6 and 8 were merged into a new category labeled "Sea." The category details and corresponding sample numbers are listed in Fig. 7 and Table II.

TABLE I
Land Cover Classes and Numbers of Samples in Chikusei

Class	Name	Pixels	Color
1	Water	2345	
2	Bare soil(school)	2859	
3	Bare soil(park)	236	
4	Bare soil(farmland)	48525	
5	Natural plants	4297	
6	Weeds in farmland	1108	
7	Forest	20516	
8	Grass	6515	
9	Rice field(groim)	13369	
10	Rice field(first stage)	1268	
11	Row crops	5961	
12	Plastic house	2193	
13	Manmade(nan-dark)	1220	
14	Manmade(dark)	7664	
15	Manmade(blue)	431	
16	Manmade(red)	222	
17	Manmade grass	1040	
18	Asphalt	801	
19	Paved ground	145	
Total		77592	

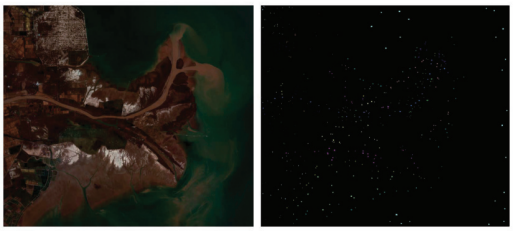


Fig. 7. Pseudocolor composite image and ground-truth map of HHK.

3) Yancheng (YC): This dataset was acquired by the ZY1-02D sensor over the Yancheng Coastal Wetlands in 2020. It consists of 166 spectral bands, with 19 noisy bands removed, leaving 147 usable bands. The spatial size of the image is 1398 × 942, with a spatial resolution of 30 m. The dataset primarily contains 16 categories. Detailed information is provided in Fig. 8 and Table III.

4) Hangzhouwan (HZW): The HZW dataset is an HSI of the Ningbo Hangzhou Bay National Wetland Park, captured by the ZY-02D sensor in 2022. It contains 166 spectral bands with 138 usable bands after removing noisy bands. The image

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TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN HHK

Class	Name	Pixels	Color
1	Salt marsh	393	
2	Acquaculture	796	
3	Mud flat	110	
4	Rice	190	
5	Aquatic vegetation	83	
6	Sea	307	
7	Freshwater marsh	95	
8	Reed	200	
9	Pond	936	
10	Build up	553	
11	Suaeda salsa	469	
12	Flood plain	361	
13	River	240	
14	Soybean	595	
15	Broomcorn	454	
16	Maize	133	
17	Locust	377	
18	Spartina	68	
Total		6360	

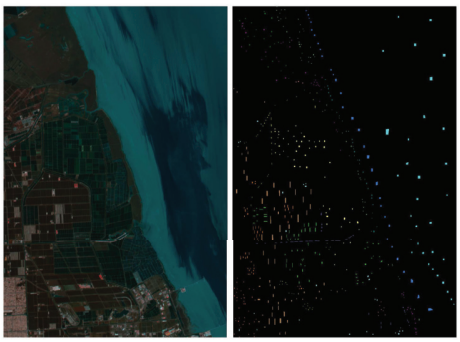


Fig. 8. Pseudocolor composite image and ground-truth map of YC.

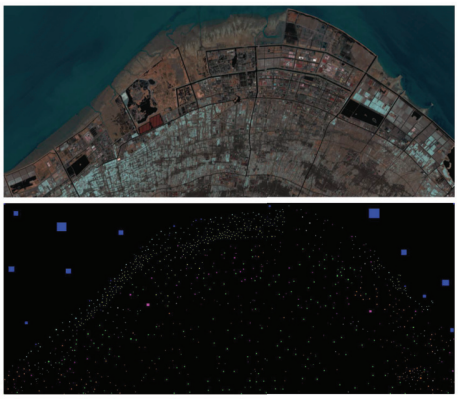


Fig. 9. Pseudocolor composite image and ground-truth map of HZW.

size is 1651 × 701, with a spatial resolution of 30 m. Seven major categories were selected for classification. Detailed information is presented in Fig. 9 and Table IV.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN YC

Class	Name	Pixels	Color
1	Architecture	451	
2	Suaeda	269	
3	River	214	
4	Beach	331	
5	Bare land	235	
6	Paddy	3026	
7	Fallow land	650	
8	Fish pond	920	
9	Sea	2076	
10	Dry land	145	
11	Salt pond	901	
12	Reed	244	
13	Offshore water	1558	
14	Pond	68	
15	Spartina	306	
16	Ditch	328	
Total		11722	

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN HZW

Class	Name	Pixels	Color
1	Ocean	5097	
2	Bare flat	588	
3	Wetland vegetation	1353	
4	Water	874	
5	Buildings	1582	
6	Bare land	1089	
7	Cultivated land	981	
Total		11564	

B. Experimental Setting

We compare the proposed method with two deep learning HSIC methods (i.e., SSRN [23] and SPFormer [41]) and six latest cross-domain FSL methods, including DCFSL [37], Gia-CFSL [38], RPCL-FSL [39], FSCF-SSL [40], ADAFSL [48], and CDFS-CASCL [49]. For SSRN and SPFormer, no source training is involved and the classification is directly performed on the target domain with only five labeled samples per class for training. Cross-domain FSL methods require training on labeled samples from the source domain. For DCFSL, Gia-CFSL, RPCL-FSL, ADAFSL, and CDFS-CASCL methods, the CH dataset is used as the source domain, with 200 labeled samples per class randomly selected for training. For FSCF-SSL, the mini-ImageNet dataset is chosen as the source domain for training [40].

In this study, we adopt an episode training strategy, constructing N-way K-shot classification tasks on the support set, where N is the number of categories in the target domain and

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TABLE V
CLASSIFICATION RESULTS (%) ON THE HHK DATASET

Class	SSRN	DCFSL	Gia-CFSL	RPCL-FSL	FSCF-SSL	ADAFSL	SPFormer	CDFS-CASCL	LGSAnet
1	63.51	77.06	84.15	85.31	81.65	90.12	89.17	83.68	**91.63**
2	89.06	94.73	93.22	92.29	92.21	99.66	91.87	99.28	**99.90**
3	64.09	**98.10**	79.52	92.86	57.62	95.87	90.48	92.38	93.52
4	74.62	80.54	87.57	**90.81**	87.30	88.83	78.37	86.85	89.08
5	79.48	91.03	71.15	83.97	85.26	69.66	85.38	92.31	**93.85**
6	89.61	91.72	90.89	98.68	91.39	95.81	97.66	98.69	**98.74**
7	79.87	90.00	99.44	**100.0**	97.22	96.67	90.04	**100.0**	97.33
8	96.04	99.49	**100.0**	99.74	**100.0**	98.29	**100.0**	**100.0**	99.28
9	83.71	82.26	87.39	86.68	**90.13**	72.29	60.47	86.65	88.58
10	87.75	99.82	98.72	96.99	99.00	94.46	93.24	97.68	**99.96**
11	89.86	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
12	93.49	97.75	94.80	94.80	99.17	97.66	89.61	98.50	**99.61**
13	59.33	72.77	61.91	**73.19**	57.45	77.87	57.02	81.42	72.17
14	99.38	99.83	98.47	98.56	**100.0**	99.49	92.54	95.93	98.95
15	91.67	98.73	96.99	95.21	96.99	94.58	91.98	99.63	**99.91**
16	56.53	72.66	90.62	85.16	87.89	86.98	89.06	95.31	**97.97**
17	75.81	80.91	72.18	**82.26**	72.18	81.09	66.51	72.76	74.68
18	63.65	80.95	**100.0**	92.86	77.78	95.77	92.06	90.48	94.60
OA	80.48 ±1.82	90.94 ±2.68	91.60 ±1.72	92.01 ±2.64	90.57 ±1.73	90.84 ±2.02	91.34 ±2.42	92.90 ±2.08	**94.38 ±1.59**
AA	79.75 ±1.25	89.74 ±1.94	89.63 ±3.02	91.63 ±2.32	87.55 ±2.11	89.59 ±0.74	89.73 ±1.39	92.78 ±1.04	**93.31 ±0.68**
$\kappa \times 100$	78.83 ±2.08	90.15 ±2.83	90.85 ±2.95	91.31 ±1.68	89.73 ±1.59	90.06 ±2.17	90.42 ±2.66	92.27 ±2.24	**93.87 ±1.15**

K is set to 1. Each query set contains 19 labeled samples per class.

All the experiments were run with the Inter Xeon CPU, NVIDIA GeForce RTX 2080Ti with 12-GB memory, and 128-GB RAM based on PyTorch 1.8.1 and CUDA 10.1 framework. Both the visual encoder and textual encoder use the Adam optimizer with a learning rate of 0.001 and a weight decay of 0.0001. The textual encoder is initialized with pretrained weights from the CLIP's ViT-B-32.pt model [44]. The evaluation metrics include overall classification accuracy (OA), average classification accuracy (AA), and the Kappa coefficient (κ). All the experiments are conducted with ten independent runs, and the final results are reported as the average of these ten runs.

C. Experimental Results

Tables V–VII present the classification results of different algorithms on three datasets, with the best results in each group highlighted in bold.

Table V presents the classification results of various methods on the HHK dataset. From the table, it is clear that our LGSAnet method consistently outperforms other methods in terms of OA, AA, and κ. Compared with state-of-the-art FSL methods such as CDFS-CASCL and ADAFSL, LGSAnet achieves more than 1.48% improvement in OA. Among the 18 land cover classes, LGSAnet delivers the highest accuracy in nine classes. Notably, LGSAnet demonstrates superior performance in classes with scarce labeled samples. For instance, in Class 5 "Aquatic vegetation," LGSAnet surpasses other methods by at least 1.64%. Moreover, for Class 15 "Broomcorn" and Class 16 "Maize," which belong to the same grass family and exhibit similar spectral profiles, most baseline methods struggle to distinguish them accurately. These methods often show better performance on Class 15 while performing poorly on Class 16. In contrast, our LGSAnet achieves consistent good performance on both the classes. These results highlight the effectiveness of LGSAnet in capturing subtle spectral differences between semantically similar categories, which can be attributed to its robust prototype alignment and the similarity-aware calibration mechanism.

Table VI presents the classification results on the YC dataset. As shown in the table, LGSAnet consistently outperforms all other methods, achieving an OA of 94.50%, an AA of 93.05%, and κ of 93.64%. LGSAnet performs well in categories with fewer labeled samples, such as Class 12 "Reed," Class 14 "Pond," and Class 16 "Ditch." In addition, LGSAnet also excels in distinguishing spectrally similar classes. For example, although Class 8 "Fish pond" and Class 14 "Pond" share highly similar spectral signatures, LGSAnet achieves an OA of 81.20% for Class 8 and 100% for Class 14, outperforming all other methods.

Table VII presents the classification results of various methods on the HZW dataset. The results clearly indicate that LGSAnet consistently outperforms all the competing methods in terms of OA, AA, and κ. Compared with other methods, LGSAnet maintains a good balance between Class 2 "Bare flat" and Class 6 "Bare land," which have limited labeled samples and are difficult to be distinguished. While existing methods tend to perform well on Class 2 at the expense of degraded performance on Class 6, LGSAnet delivers robust results on both the classes. This highlights its capability to maintain discriminative performance even under data-scarce and ambiguous class conditions.

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TABLE VI
CLASSIFICATION RESULTS (%) ON THE YC DATASET

Class	SSRN	DCFSL	Gia-CFSL	RPCL-FSL	FSCF-SSL	ADAFSL	SPFormer	CDFS-CASCL	LGSAnet
1	**90.27**	78.70	87.89	77.58	76.91	86.55	88.12	75.93	83.74
2	71.18	**92.80**	92.05	87.12	83.71	92.55	62.12	91.67	85.98
3	63.28	94.50	94.02	**96.65**	94.98	92.82	95.69	91.07	95.22
4	65.76	97.24	96.17	95.86	93.25	95.50	95.71	90.59	**100.0**
5	91.73	99.78	99.78	99.13	**100.0**	99.42	**100.0**	99.28	99.57
6	93.74	99.30	99.77	**100.0**	94.99	93.43	95.46	95.99	96.99
7	91.88	95.74	94.11	92.79	97.05	93.44	96.37	95.04	**97.13**
8	70.45	66.94	67.87	74.92	73.88	60.98	80.18	75.74	**81.20**
9	93.35	99.98	**100.0**	98.62	**100.0**	86.91	**100.0**	99.15	98.55
10	73.66	**98.57**	92.14	90.69	95.66	93.57	82.43	88.81	90.71
11	81.17	93.69	**94.98**	93.02	94.21	90.96	94.97	85.64	93.47
12	76.29	97.70	96.65	95.82	92.26	**99.30**	80.33	97.21	98.95
13	93.75	85.96	82.55	84.93	94.07	92.44	97.03	93.30	**97.74**
14	67.24	97.62	98.27	99.21	94.44	96.30	**100.0**	98.94	**100.0**
15	80.71	66.45	66.61	66.45	83.55	**98.78**	84.73	93.24	92.69
16	63.44	71.83	71.05	72.29	75.10	66.46	68.11	73.89	**76.93**
OA	82.37 ±3.76	91.74 ±2.59	91.68 ±1.63	91.66 ±1.46	92.36 ±1.38	88.81 ±3.11	92.76 ±2.67	91.98 ±3.64	**94.50 ±1.24**
AA	77.39 ±4.07	89.80 ±2.02	89.68 ±2.70	89.07 ±1.66	90.12 ±1.98	89.96 ±1.09	89.21 ±1.77	90.34 ±4.00	**93.05 ±1.20**
$\kappa \times 100$	80.92 ±4.11	90.44 ±2.72	90.37 ±1.88	90.35 ±2.39	91.33 ±1.59	87.16 ±3.53	91.63 ±3.05	90.76 ±4.17	**93.64 ±1.43**

TABLE VII
CLASSIFICATION RESULTS (%) ON THE HZW DATASET

Class	SSRN	DCFSL	Gia-CFSL	RPCL-FSL	FSCF-SSL	ADAFSL	SPFormer	CDFS-CASCL	LGSAnet
1	95.17	**99.73**	99.69	99.04	99.68	99.63	94.44	96.57	**99.73**
2	58.47	77.02	81.90	93.83	90.82	93.48	86.51	**94.34**	79.38
3	78.11	63.43	78.04	55.79	66.06	67.11	76.38	79.03	**80.21**
4	80.42	98.04	92.52	**99.42**	95.05	96.16	92.28	92.48	95.37
5	87.91	74.51	73.30	90.23	**91.50**	84.99	82.88	85.46	88.46
6	48.19	69.48	57.98	60.38	48.15	46.89	56.38	60.30	**70.12**
7	78.28	88.93	90.27	79.15	79.56	80.26	89.37	89.41	**90.75**
OA	75.14 ±1.65	87.09 ±2.15	87.42 ±1.78	87.22 ±1.57	87.28 ±1.41	86.66 ±1.99	86.35 ±2.43	88.61 ±1.87	**89.96 ±1.12**
AA	67.31 ±2.29	81.73 ±1.69	82.24 ±1.25	82.55 ±2.49	81.55 ±1.64	81.22 ±3.02	80.41 ±3.25	84.76 ±1.53	**85.13 ±1.09**
$\kappa \times 100$	66.47 ±1.94	82.76 ±2.28	83.49 ±2.67	83.00 ±1.71	82.99 ±1.52	82.23 ±2.63	81.54 ±3.11	85.12 ±1.79	**86.05 ±1.46**

Since the available labels in a full scene are relatively sparse, we first generate a full-scene classification map for each target dataset using all the available labeled samples. Specifically, we train a HybridSN model on all the available labeled samples from the target domain and classify the entire scene to produce a full classification map, which is called as the "Oracle" classification map. Figs. 10–12 visually demonstrate the classification map of different methods on the three datasets. It can be observed that the classification maps produced by our method are much closer to the "Oracle."

D. Parameter Analysis

Here, we analyze the effect of parameter α in the SAPC module. Fig. 13 shows the OA of LGSAnet versus the parameter α on three datasets, where α is chosen from 0.2, 0.4, 0.6, 0.8, and 1. It can be seen that the optimal α is 0.8 for the HHK and HZW datasets and 0.6 for the YC dataset.

This difference may be due to the domain shift difference between the source and target domains. We use the maximum mean discrepancy (MMD) to measure the distribution difference between domains. The MMD values of the HHK, YC, and HZW datasets and the source domain are 0.26, 0.12, and 0.23, respectively. The HHK and HZW datasets have a large distribution difference with the source domain, and a larger α can reduce the risk of propagating irrelevant source domain knowledge. In contrast, the YC dataset is more aligned with the source domain features, and a smaller α can effectively balance the knowledge transfer from the source to the target. To ensure consistency, α is set to 0.8 in the experiments.

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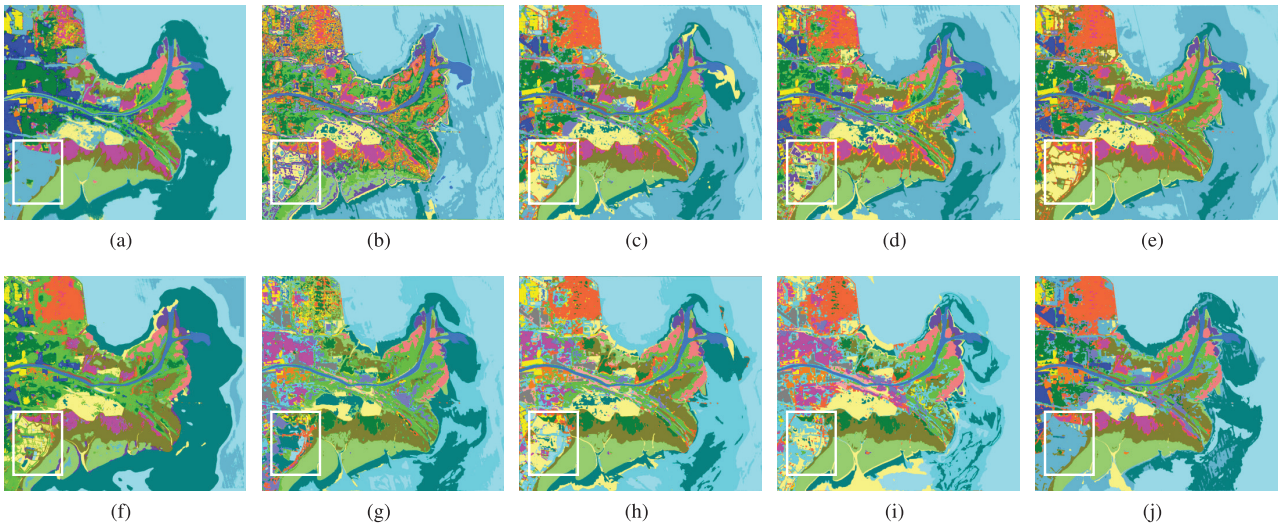


Fig. 10. Classificationmap on HHK. (a) Oracle. (b) SSRN. (c) DCFSL. (d) Gia-CFSL. (e) RPCL-FSL. (f) FSCF-SSL. (g) ADAFSL. (h) SPFormer. (i) CDFS-CASCL. (j) LGSAnet.

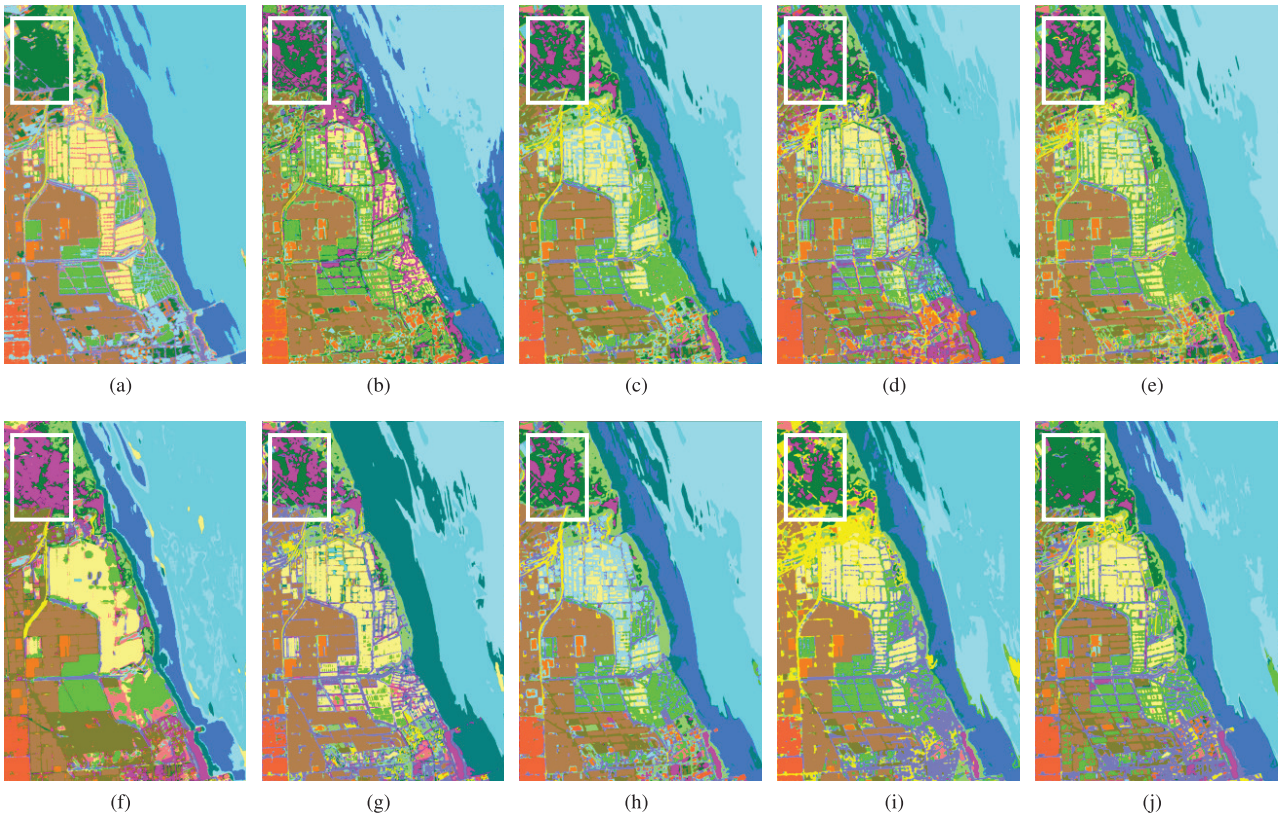


Fig. 11. Classificationmap on YC. (a) Oracle. (b) SSRN. (c) DCFSL. (d) Gia-CFSL. (e) RPCL-FSL. (f) FSCF-SSL. (g) ADAFSL. (h) SPFormer. (i) CDFS-CASCL. (j) LGSAnet.

We further analyze the influence of the temperature coefficient τ in the contrastive loss on three datasets as shown in Fig. 14. It can be seen that $\tau = 0.4$ yields the best performance on the HHK and HZW datasets, while $\tau = 0.5$ achieves the highest accuracy on the YC dataset. Nevertheless, $\tau = 0.5$ is widely used in prior contrastive learning studies [51], [52] as a robust and empirically choice. Therefore, we adopt $\tau = 0.5$ as the default setting in all the experiments to maintain consistency.

E. Statistical Analysis

To validate whether the differences in classification performance between LGSAnet and other compared methods are statistically significant, we perform the McNemar's test for each method. The McNemar's test [53] computes the Z-statistic as follows:

$$Z = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}} \tag{15}$$

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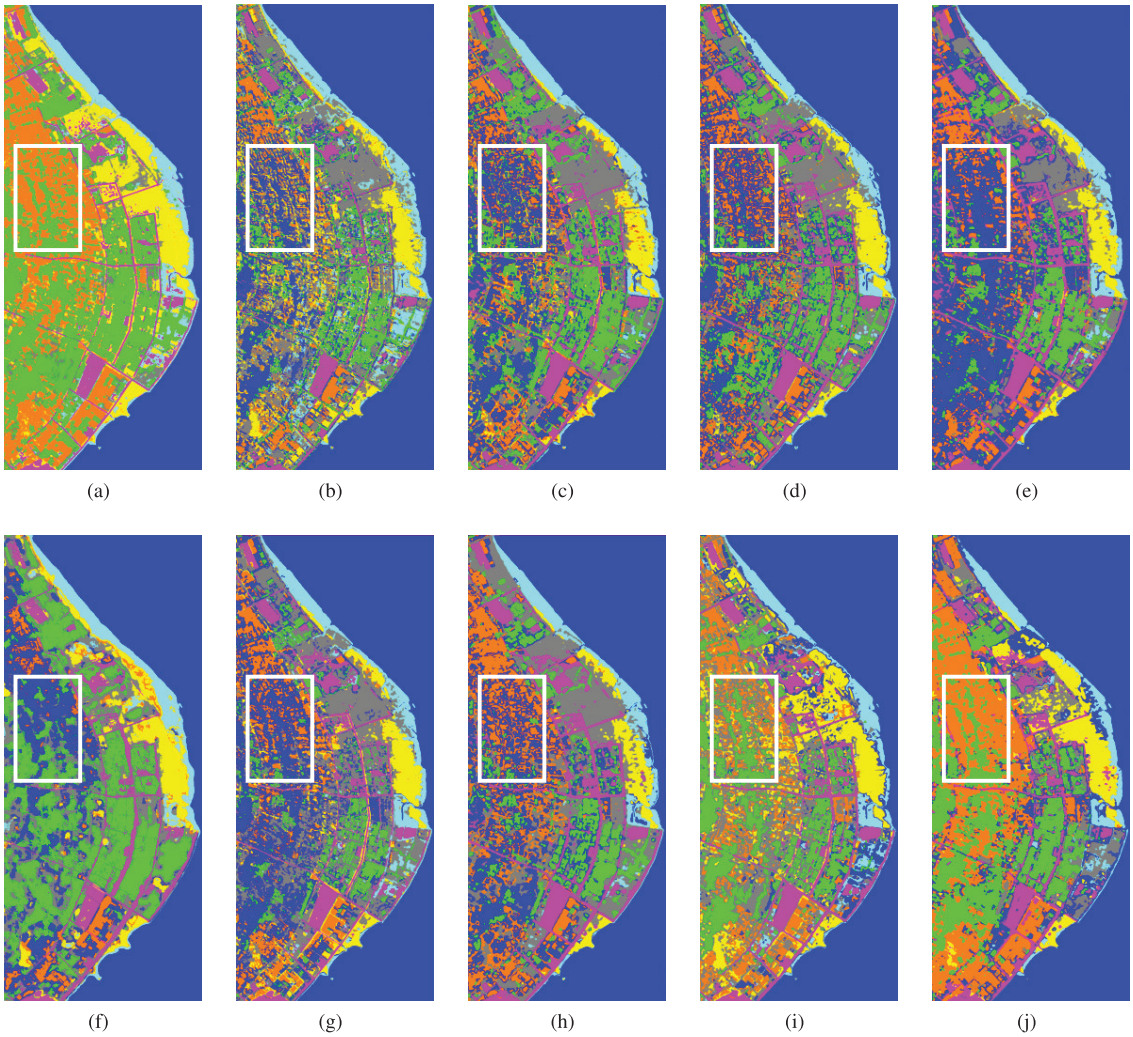


Fig. 12. Classificationmap on HZW. (a) Oracle. (b) SSRN. (c) DCFSL. (d) Gia-CFSL. (e) RPCL-FSL. (f) FSCF-SSL. (g) ADAFSL. (h) SPFormer. (i) CDFS-CASCL. (j) LGSAnet.

TABLE VIII

McNema's Test on Three Datasets to Validate Whether the Difference Between LGSAnet and Each of Other Methods is Statistically Significant

		SSRN	DCFSL	Gia-CFSL	RPCL-FSL	FSCF-SSL	ADAFSL	SPFormer	CDFS-CASCL
HHK	OA	80.48	90.94	91.60	92.01	90.57	90.84	91.34	92.90
	Z	−19.21	−7.36	−6.86	−6.57	−8.27	−7.92	−7.08	−5.78
YC	OA	82.37	91.74	91.68	91.66	92.36	88.81	92.76	91.98
	Z	−23.84	−12.08	−12.34	−12.79	−11.12	−16.37	−10.51	−11.93
HZW	OA	75.14	87.09	87.42	87.22	87.28	86.66	86.35	88.61
	Z	−25.89	−9.68	−9.24	−9.43	−9.41	−10.04	−10.39	−4.81

where f_{12} denotes the number of test samples misclassified by LGSAnet but correctly classified by the compared method, while f_{21} indicates the number of samples misclassified by the compared method but correctly classified by LGSAnet [53]. Following the conventional 5% significance level, the difference between LGSAnet and each method is considered statistically significant if $|Z| > 1.96$ [53]. A negative Z-value suggests that LGSAnet is statistically superior to the compared method, while a positive value implies the opposite.

Table VIII presents the statistical significance results using McNemar's test on the HHK, YC, and HZW datasets. For all the three datasets, the Z-scores are consistently less than −1.96, indicating that the performance gains are statistically significant at the 5% confidence level.

F. Ablation Analysis

The proposed LGSAnet mainly consists of three modules: FSL, LGPA, and SAPC. To validate the effectiveness of each

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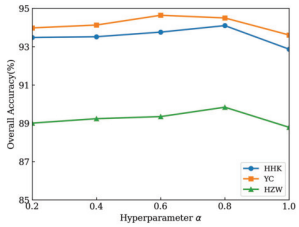


Fig. 13. OA of LGSAnet versus different parameter α.

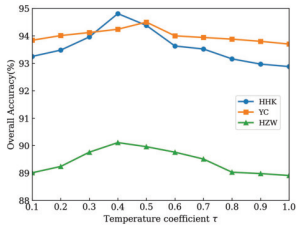


Fig. 14. OA of LGSAnet versus different parameter τ.

TABLE IX
OA OF DIFFERENT MODULES ON THREE DATASETS

Modules			Datasets		
FSL	LGPA	SAPC	HHK	YC	HZW
✓			91.23	92.68	87.91
✓	✓		92.87	93.61	88.79
✓		✓	93.48	93.97	89.36
✓	✓	✓	94.38	94.50	89.96

module, we perform ablation experiments on three datasets, as shown in Table IX. We consider the following four variants.

1) Only performing FSL on the source and target domains.
2) Only adding the LGPA module to the FSL framework.
3) Only adding the SAPC module to the FSL framework.
4) Adding both the LGPA and SAPC modules to the FSL framework.

From Table IX, we can see that after adding the LGPA module, the OA on the HHK, YC, and HZW datasets increased by 1.64%, 0.93%, and 0.88%, respectively. Similarly, adding the SAPC module alone results in OA improvements of 2.25%, 1.29%, and 1.45% on these datasets. It is clear that all the modules contribute to the classification, and the integration of all the modules yields the best overall performance.

To visually evaluate the effectiveness of the LGPA module, we select five categories from each dataset and use the t-distributed stochastic neighbor embedding (t-SNE) strategy to visualize the extracted features in Fig. 15. It can be seen that the LGPA module significantly improves the intraclass compactness, while also enhancing interclass separation. This improvement may be attributed to the fact that the LGPA module effectively combines visual and textual modal information, enabling the prototype to better capture the semantic characteristics of the category.

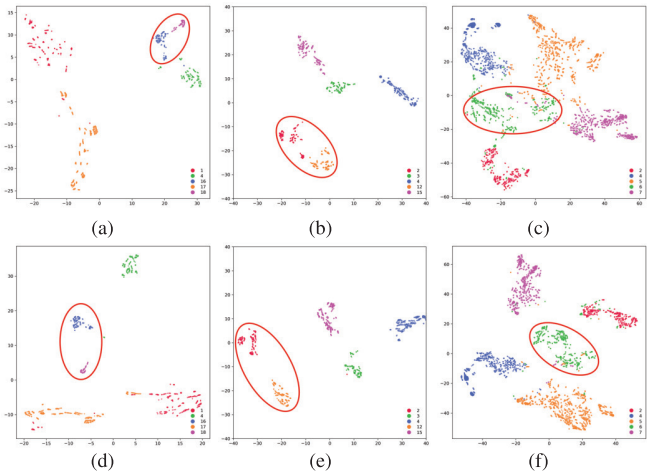


Fig. 15. Feature visualization of the LGPA module using the t-SNE on three datasets. (a) General FSL on HHK (S = 7.72). (b) General FSL on YC (S = 5.41). (c) General FSL on HZW (S = 4.28). (d) Adding LGPA to FSL on HHK (S = 8.22). (e) Adding LGPA to FSL on YC (S = 7.65). (f) Adding LGPA to FSL on HZW (S = 5.62).

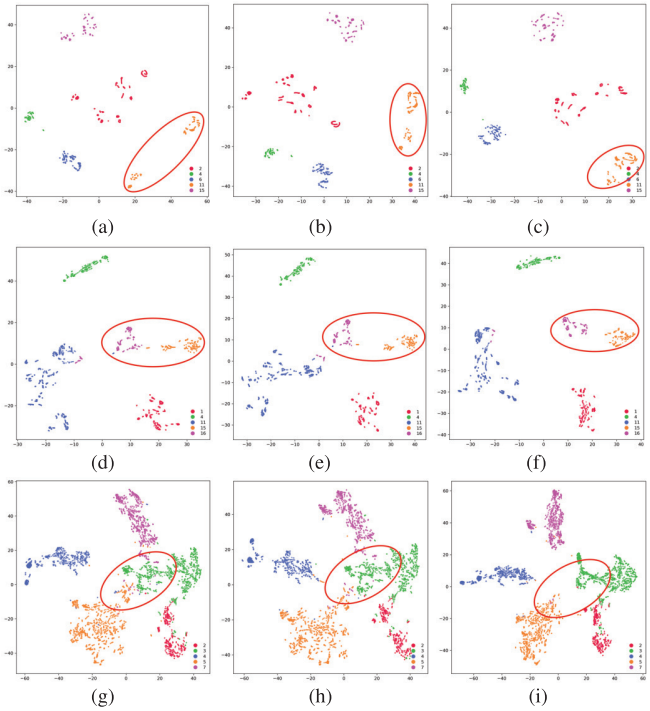
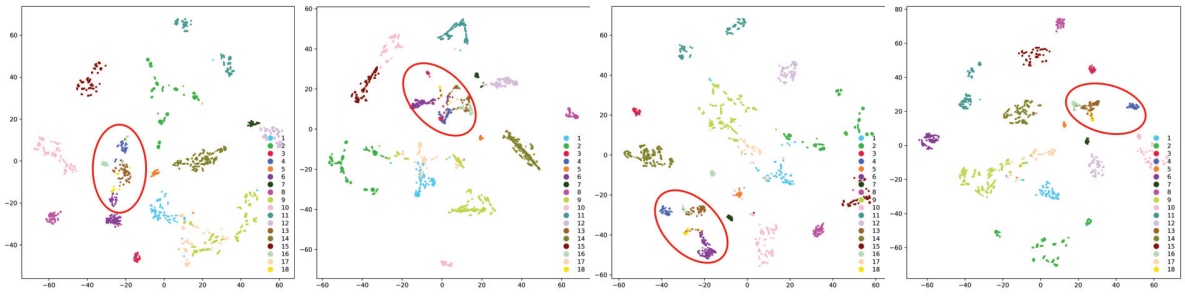


Fig. 16. Feature visualization of the SAPC module using the t-SNE on three datasets. (a) General FSL baseline on HHK (S = 7.07). (b) FSL with MMD on HHK (S = 8.59). (c) FSL with SAPC on HHK (S = 10.02). (d) General FSL baseline on YC (S = 5.03). (e) FSL with MMD on YC (S = 5.72). (f) FSL with SAPC on YC (S = 6.75). (g) General FSL baseline on HZW (S = 4.66). (h) FSL with MMD on HZW (S = 4.87). (i) FSL with SAPC on HZW (S = 5.42).

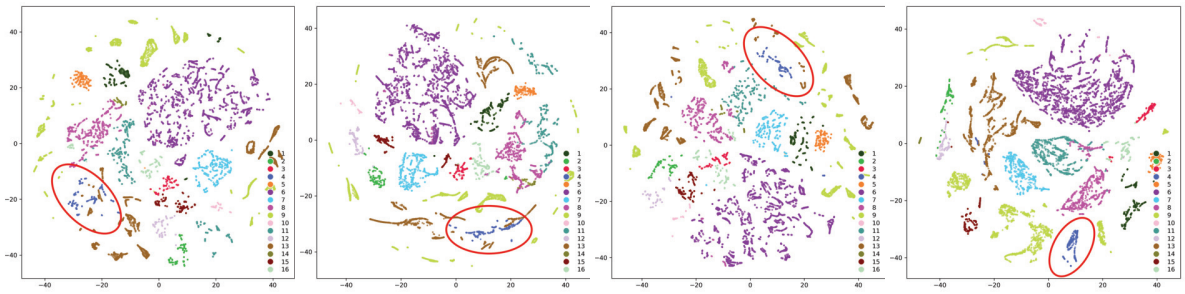
To further illustrate the effectiveness of the proposed SAPC module, we visualize the prototype distributions before and after calibration. As shown in Fig. 16, SAPC significantly enhances the compactness of prototypes. To highlight its superiority, we also compare SAPC with traditional domain adaptation methods such as MMD. The results demonstrate

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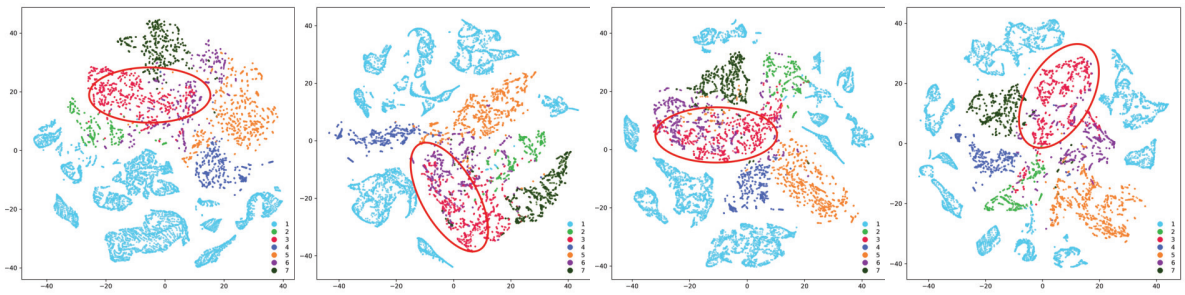
(a) (b) (c) (d)

Fig. 17. Feature visualization of different methods using the t-SNE on HHK. (a) DCFSL (S = 4.11). (b) Gia-CFSL (S = 4.93). (c) RPCL-FSL (S = 5.85). (d) LGSAnet (S = 8.94).



(a) (b) (c) (d)

Fig. 18. Feature visualization of different methods using the t-SNE on YC. (a) DCFSL (S = 4.04). (b) Gia-CFSL (S = 6.81). (c) RPCL-FSL (S = 5.17). (d) LGSAnet (S = 8.29).



(a) (b) (c) (d)

Fig. 19. Feature visualization of different methods using t-SNE on HZW. (a) DCFSL (S = 1.90). (b) Gia-CFSL (S = 2.33). (c) RPCL-FSL (S = 3.46). (d) LGSAnet (S = 4.58).

that SAPC achieves better interdomain prototype alignment and stronger intraclass compactness than MMD.

G. Feature Separability

To visually evaluate the feature extraction capabilities of different methods, we use the t-SNE strategy to visualize the features extracted by four methods: DCFSL, Gia-CFSL, RPCL-FSL, and LGSAnet. As illustrated in Figs. 17–19, the features extracted by the proposed LGSAnet are more distinctly separated across different categories, indicating stronger discriminative ability. In contrast, the features extracted by other methods tend to overlap or cluster together.

To quantitatively assess the discriminative power of the learned feature representations, we introduce the Fisher discriminant score (FDS) to measure class separability as follows:

$$S = \frac{\text{trace}\,(S_b)}{\text{trace}\,(S_w)} \tag{16}$$

where S_b and S_w denote the between-class and within-class scatter matrices, respectively. The FDS values of features in Figs. 15–19 are computed. A higher FDS value S indicates stronger discriminative ability of the feature representations.

H. Computational Complexity

Table X presents the computational complexity of various methods, where the training time, testing time, floating point operations (FLOPs), and the number of parameters are considered. Compared with most cross-domain FSL-based methods (e.g., DCFSL, RPCL-FSL, FSCF-SSL, and ADAFSL), our proposed LGSAnet has fewer parameters, and its computation time and FLOPS are lower than those of Gia-CFSL,

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TABLE X
COMPUTATIONAL COMPLEXITY ON THREE DATASETS

Dataset	Metric	Methods							
		DCFSL	Gia-CFSL	RPCL-FSL	FSCF-SSL	ADAFSL	SPFormer	CDFS-CASCL	LGSAnet
HHK	Training time	2263.18	6611.45	1025.42	3332.13	3516.08	1960.22	2016.13	1910.72
	Testing time	1.59	1.43	2.43	9.18	1.52	4.83	1.04	3.07
	FLOPS	44.95	26814.36	44.95	1031.27	43.84	1389.36	255.73	273.64
	#Params	4.06	0.85	4.26	6.19	4.25	0.87	0.93	1.39
YC	Training time	1814.15	5612.65	932.41	2859.07	2885.52	1521.04	1714.32	1498.34
	Testing time	2.21	2.57	3.48	10.58	2.19	7.27	1.57	4.39
	FLOPS	43.72	26811.78	43.72	1025.48	43.52	1158.55	253.29	285.42
	#Params	4.07	0.86	4.27	6.23	4.25	0.85	0.91	1.41
HZW	Training time	1106.76	3721.43	646.21	1367.09	1802.06	1245.59	1558.02	826.17
	Testing time	1.98	2.39	3.98	9.67	2.11	5.75	1.56	4.24
	FLOPS	43.72	26812.51	43.72	1025.48	43.18	504.62	252.89	281.42
	#Params	4.07	0.86	4.27	6.21	4.24	0.85	0.91	1.41

FSCF-SSL, and SPFormer. Although the training time is higher than that of RPCL-FSL, the classification performance of LGSAnet shows a significant improvement. Overall, our LGSAnet method achieves better classification performance at an acceptable computational cost.

IV. CONCLUSION

In this article, we have proposed an LGSAnet for few-shot coastal wetland HSIC. The proposed LGSAnet mainly consists of an LGPA module and an SAPC module. The LGPA module is designed to guide the learning of visual prototypes through linguistic features and to improve the semantic representation ability of prototypes. The SAPC module is used to calibrate target domain prototypes by leveraging the similarity between the source and target domain prototypes, which can mitigate domain shift. By improving the prototype learning through language-guided alignment and similarity-aware calibration, the proposed LGSAnet shows good performance in cross-domain few-shot HSIC tasks.

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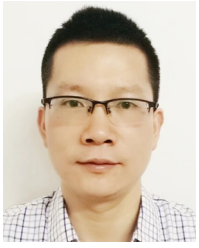
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